1. Selection & Assembly Strategy

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🚀 **Chosen Model:** Arcturus. Selected for optimal long internal volume perfectly suited for a custom electronics sled.

🚀 **Aerodynamic Advantage:** Shifting the avionics mass forward relocates the Center of Gravity, maximizing flight stability.

🚀 **Assembly & Access:** Reinforced bonding at ailerons & shock cord. Friction-fit, removable nose cone ensures post-assembly access.

2. Electronics Bay Integration

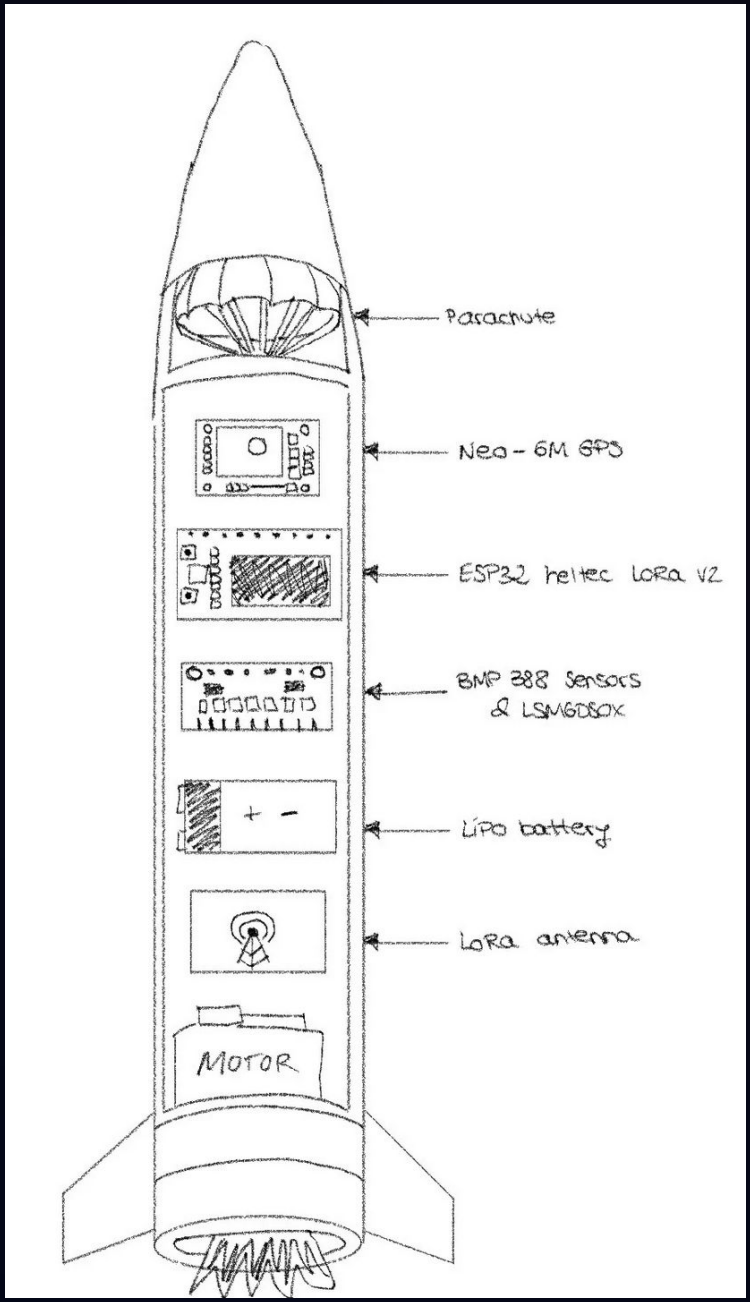
Internal Payload Architecture

🚀 **Payload Vertical Stack:** NEO-6M GPS (Top), ESP32 Heltec LoRa, Sensors (BMP388 & LSM6DSOX), LiPo Battery, LoRa Antenna.

🚀 **Mechanical Survival:** Components firmly secured to the internal sled to prevent shift during launch forces.

🚀 **Vibration Mitigation:** High-frequency thrust vibrations are absorbed using high-density mounting foam.

🚀 **System Safety:** Complete physical isolation between avionics and recovery compartment prevents parachute interference.



3. Motor Constraints & Conclusion

The "-6" Delay

Mandatory **D12-6 Motor (24mm)** dictates a 6-second coast phase before ejection. Critical to allow the rocket to reach apogee, preventing high-speed deployment destruction.

Ejection Risks

Mechanical shock and thermal stress from ejection gases are mitigated by the internal wooden baffle and our physical sled barrier. Mass contingency plan: E20 motor upgrade if needed.

FINAL RECOMMENDATION

"Our recommended rocket solution is the Arcturus because it provides the optimal internal volume to safely house our custom electronics bay, ensures excellent aerodynamic stability by shifting the center of gravity forward, and allows for secure physical separation between our avionics and the parachute ejection system."